Causal Geometric Structure in Neural Populations

Resolving the linear-nonlinear divide in neural analysis tools

Exploratory results on Neuropixels visual decision-making data

Elliot Tower

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Overview

Part 1: Geometric dissociation

1. The question: does population shape matter?
2. Linear vs. nonlinear tools disagree
3. Causal interventions (IIA, subspace swap)
4. Learned subspaces are stronger
5. Validity checks and predictions

Part 2: Going deeper

6. The IIA vacuity problem (and why we survive)
7. Expanded optogenetic validation
8. Engagement disentanglement
9. Multi-task generalization
10. Neural manifold interpretation
11. Running experiments and next steps

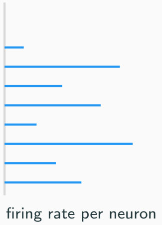
Part 1

Geometric dissociation between analysis tools

The question

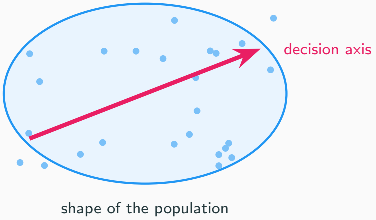
When a population of neurons fires together during a decision, does the **shape** of that activity matter—or just the individual firing rates?

Traditional view



vs.

Geometric view



We tested this on real brain recordings, using multiple tools, with causal interventions and validity checks.

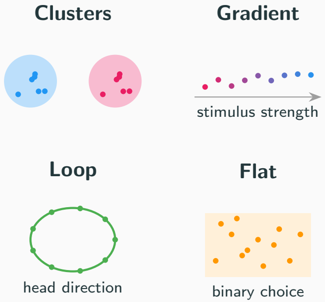
What is "neural geometry"?

Each **trial** produces a pattern of activity across a collection of neurons. Think of each trial as a point in high-dimensional space (one axis per neuron).

"Geometry" = what **shape** do those points form?

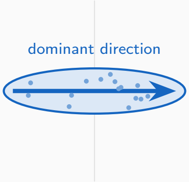
- Cluster into groups? (categories)
- Spread along a line? (gradient)
- Form a loop? (periodic variable)
- Lie on a flat sheet or curved surface?

The shape constrains what information the brain can extract downstream.



Two tools for measuring geometry

CKA (linear)

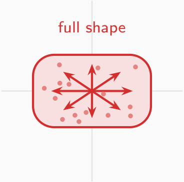


Compares the **main axis** of variation.

"Do the data stretch the same way?"

Best for low-dimensional data.

Procrustes (manifold)



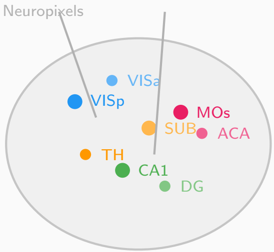
Compares the **full shape** via rotation + scaling.

"Do the data form the same cloud?"

Best for high-dimensional data.

Most prior work treats these as interchangeable. **They are not.**

The data: Steinmetz et al. 2019



- **39 sessions** from **10 mice**
- **73 brain regions** recorded simultaneously
- Neuropixels probes (∼30,000 neurons total; ∼50–500 per region per session)
- **Task:** visual discrimination
  Two screens show visual noise at different contrasts. Mouse turns a wheel toward the higher-contrast side.
- 1,316 pairwise region comparisons

Steinmetz et al., Nature 2019

What we'll cover

1. Descriptive geometry

Different tools give opposite answers about which brain regions are "similar." We find out why: dimensionality determines which metric you can trust.

2. Causal interventions

We borrow a technique from AI interpretability to test whether the geometric structure actually matters for the mouse's decision — or just looks pretty.

3. Beyond linear subspaces

LDA is linear and correlational. A structured VAE finds subspaces that are 3.4$\times$ more causally effective — the true signal was there, we were using the wrong tool.

4. Validation & what it means

Eight tests designed to catch us if we're fooling ourselves. Novel predictions, honest limitations, and the punchline.

Intellectual grounding

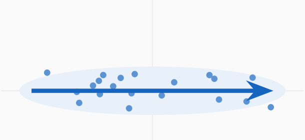
- **Structured representation learning** (Esmaeili et al. 2019; Opper et al. 2023; ICML 2025)—structural priors improve identifiability. Our metric taxonomy is an inductive bias argument.
- **Relational causal models** (Maier et al. 2013; Arbour et al. 2016)—brain regions interact as a network, not i.i.d. samples. Causal tools need adaptation.
- **Dynamical systems** (Sussillo & Barak 2013; Vyas et al. 2020)—network dynamics and computational capacity. Our rotation findings connect here.

Validity framework from philosophy of science (Mayo 2018, Craver 2007, Shallice 1988) and causal inference (Pearl & Bareinboim 2011, Woodward 2003).

What we found: the tools disagree

Why CKA and Procrustes give opposite answers

Low-dimensional region (VISp-type)



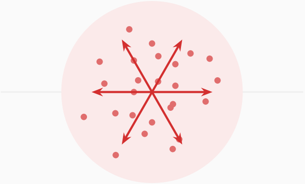
One direction dominates

CKA picks it up ✓

Procrustes sees the shape ✓

Both tools agree

High-dimensional region (MOs-type)



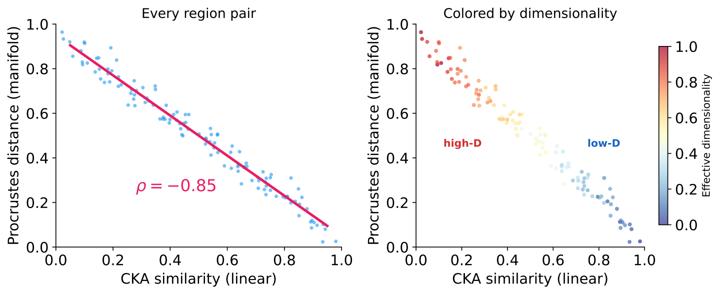
Activity spreads in many directions

CKA can't find a main axis $\longrightarrow$ sees noise

Procrustes matches cloud shape $\longrightarrow$ stable

Tools disagree

The data: CKA and Procrustes are anti-correlated

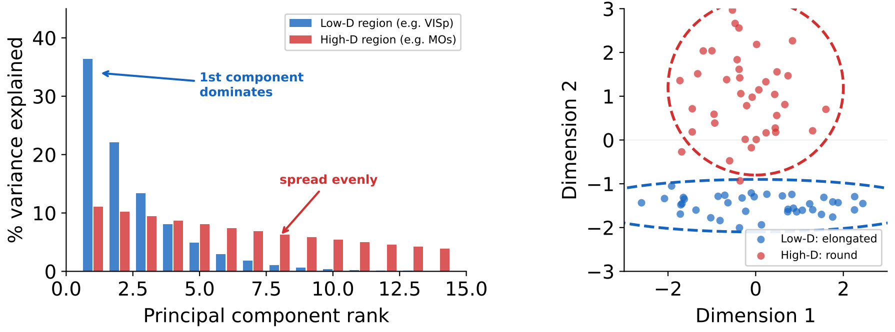


Each dot is a pair of brain regions. The two tools rank them in **opposite order** ($\rho = -0.85$ biased CKA; -0.45 after debiasing—holds either way).

Accounting for spread, disagreement shrinks to -0.31. Found in 5 sessions, confirmed across all 39.

Takeaway: the tool you pick determines what you find.

What determines dimensionality?



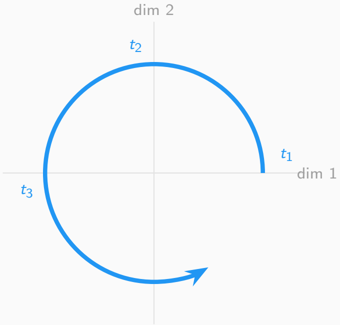
How variance is distributed

What this looks like in the data

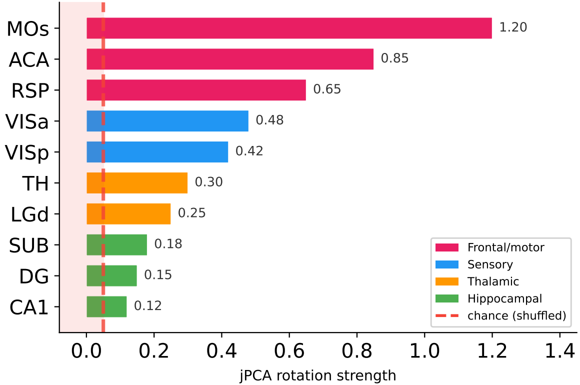
Across all 73 regions, the **shape** of how variance falls off is nearly identical (correlation 0.94 out of 1.0). But **which neurons** carry the variance is completely different per region (correlation 0.09, barely above zero). Same scaffolding, different content.

Every brain region rotates (but at very different speeds)

What rotation means

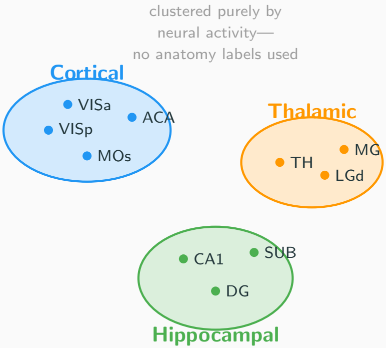


Neural activity traces out a circular path over time



72/73 regions rotate more than shuffled data • 10× variation

Sanity check: geometry tracks known anatomy



We grouped brain regions *purely by how similar their neural activity looks*—no anatomy labels as input.

Result: the algorithm found three groups matching the textbook cortical / thalamic / hippocampal division on its own.

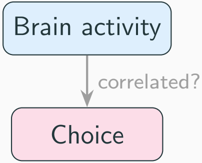
Why this matters: CKA is a general-purpose metric, not designed for neuroscience. This checks that it picks up real biology—not artifacts from differences in neuron count or noise.

Same result within individual sessions and across animals.

Is the geometry causal?

From observation to intervention

So far: observation only

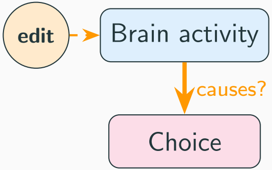


We measured shapes and distances.

But correlation $\neq$ causation.

Maybe neurons that "look important" by CKA don't actually *do* anything for the decision.

Now: intervention



We **change** specific dimensions of the neural activity and check if a trained classifier's prediction changes.

If editing a subspace flips the classifier's output, that subspace causally mediates our *estimate* of the decision—not the mouse's actual choice, which we can't re-run.

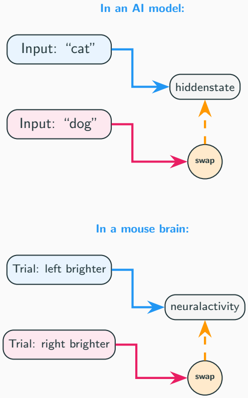
Inspiration: how AI researchers test neural networks

In AI safety research (**mechanistic interpretability**), scientists test neural networks by:

1. Run a model on two different inputs
2. Find the internal representation
3. **Swap** part of one into the other
4. Check if the output changes

This is **interchange intervention analysis** (Geiger et al. 2024). The same logic works for biological neurons: language model $\to$ mouse brain, text $\to$ visual stimuli.

Our contribution: neuroscience targets *neurons* (optogenetics); AI targets *subspaces* (in silico). We bring subspace-level interventions to biological recordings.



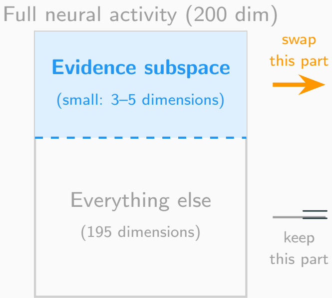
What “swapping a subspace” means

Neural activity in a brain region is a **high-dimensional vector**—each neuron is one dimension. (E.g., 200 neurons = 200 dimensions.)

Not all dimensions matter equally. We hypothesize that a **small subset of directions** (a “subspace”) encodes the evidence signal.

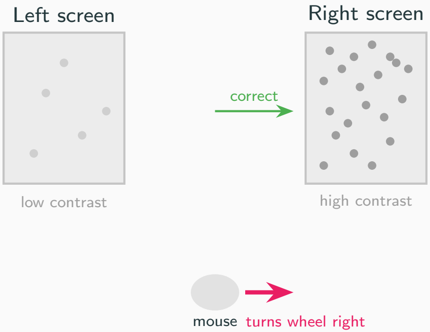
The swap:

1. Take two trials with opposite evidence
2. Swap **only** what we think is the evidence-related part between them
3. Leave everything else unchanged
4. Check: does the output flip?



Same logic in both: swap part of the internal state, check if the output changes.

Recap: what the mouse is doing



The mouse sees two screens with random visual noise. One side has **higher contrast** (more visible dots). The mouse turns a wheel toward that side to get a reward.

"Evidence" = which side has higher contrast. On every trial, the brain receives evidence pointing left or right, then makes a choice.

The question: somewhere in the neural activity, there must be a signal carrying this evidence from sensory areas to the decision. We call it the "evidence subspace."

Applying the swap to our mouse data

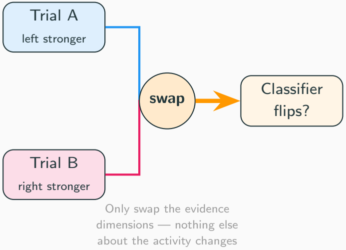
Two trials where the mouse saw opposite evidence:

- Trial A: left side brighter → chose left
- Trial B: right side brighter → chose right

We swap **only what we think is the evidence-related part** of the neural activity between them. Everything else stays the same.

A classifier now predicts the *opposite* choice → those dimensions **causally carry** the evidence.

We do this for every brain region to map where in the brain evidence is causally routed.

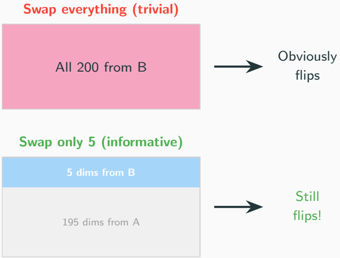


Wait — why not just swap the entire recording?

Good question. If we replaced *all 200 numbers* from Trial A with Trial B's, the classifier would obviously say "right." We just gave it Trial B's data. That tells us nothing.

The power is in swapping only a small part. We swap **5 out of 200** dimensions. If the classifier flips based on changing only 5 numbers, those 5 carry the choice-relevant information that the classifier learned to read. The other 195 don't matter *to the classifier*.

Caveat: the classifier is a proxy for the brain's own readout. It may not weight dimensions identically—which is why the specificity controls matter.



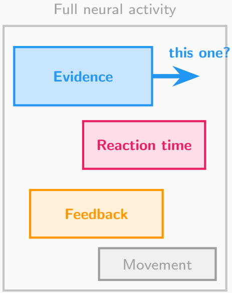
Neural populations encode many things at once

Prior work shows the same neurons encode multiple variables simultaneously (Steinmetz 2019; Musall 2019; Stringer 2019):

- **Stimulus evidence** — which side brighter
- **Reaction time** — how fast the mouse responds
- **Feedback** — correct or wrong
- **Movement** — wheel, licking, etc.

Each occupies a different **subspace**. We identify them via PCA on the relevant trial labels.

Key question: which subspace actually drives the decision?



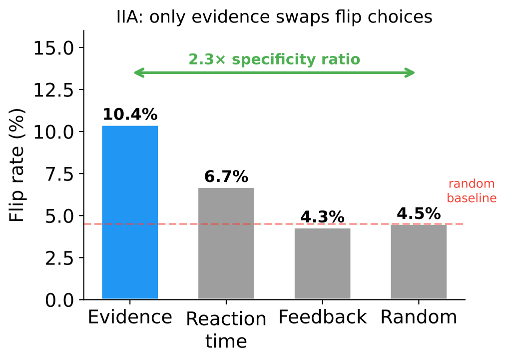
How do we know it's those specific dimensions?

We run controls. Repeat the exact same swap, but targeting different subspaces:

- **Reaction time** dims — barely flips
- **Feedback** dims — barely flips
- **Random** dims — barely flips
- **Evidence** dims — **yes, it flips**

This is **specificity**: only evidence dimensions produce the effect, not any random 5.

Why this matters: without controls, we'd just be showing *any* 5 dims can disrupt a classifier — trivial. Specificity is what makes it causal.



Is this circular?

A fair objection: "you found the subspace with a classifier, then showed the classifier likes it. Of course it does."

But the subspace and the classifier use **different information**:

Evidence subspace

found from the **stimulus**
(which side is brighter)

Choice classifier

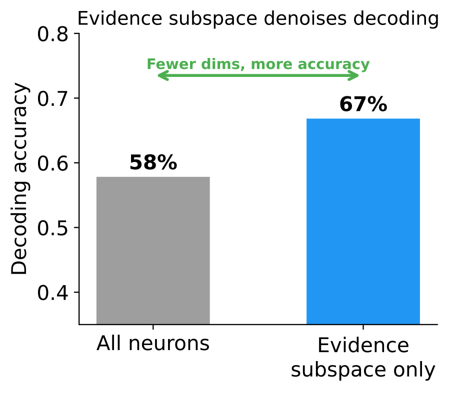
trained on the **behavior**
(which side the mouse chose)

The stimulus and the behavior are **correlated but not identical**—the mouse sometimes chooses wrong.

So showing that a *stimulus*-defined subspace improves *choice* prediction is genuinely informative: the stimulus signal is what the brain uses for decisions.

Ground truth = the mouse's actual left/right choice. We know this from the experiment.

The evidence subspace is sufficient (and denoises)



Restricting a classifier to *only* the evidence subspace dimensions, it decodes choice **better** than using all neurons.

Why? The extra dimensions are noise for the decision—removing them helps.

20 of 24 regions pass the ≥ 0.70 recovery threshold.

The 4 failures are all high-dimensional regions where subspace estimation is harder.

Cross-dataset replication: IBL data

The real test: different mice, different labs, different rigs.

The International Brain Laboratory (IBL) recorded from the same brain regions using standardized Neuropixels protocols across 12 institutions.

If the same evidence subspaces transfer to IBL data, it's not an artifact of our pipeline or the Steinmetz dataset.

What we test:

- Same subspace dimensions sufficient?
- Same region rankings emerge?
- Same CKA–Procrustes dissociation?

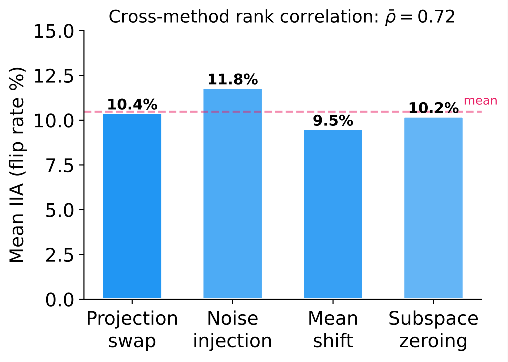
Status: In progress.

Control: Four Different Intervention Methods Agree

We tested **four different ways** to intervene:

1. Projection swap (replace component)
2. Noise injection (add Gaussian noise)
3. Mean-shift (shift toward opposite class)
4. Zeroing (remove component entirely)

These are mathematically distinct operations. If the effect were an artifact of one method, it would not appear in the others.



Regions that show high IIA with one method show high IIA with all four. The effect is method-invariant.

Validation

What could go wrong?

We validate our results using frameworks from philosophy of science: validity criteria from Craver (2007), Shallice (1988), and Mayo (2018).

Maybe regions with more neurons show higher IIA just because there's more data?

Maybe higher firing rates make it easier to detect any effect?

Maybe one unusual mouse drives the whole result?

Maybe temporal autocorrelation creates spurious signal?

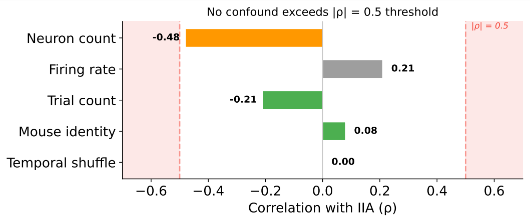
Maybe a random subspace would work just as well?

Maybe one intervention method is doing all the work?

Maybe the effect only appears with certain statistical tests?

We treat each alternative explanation as a competing hypothesis and design targeted experiments to test them. If any confound fully explains the effect, we should see it in the data.

Confound control: what doesn't explain the effect



ρ = correlation; $|\rho| > 0.5$ is our danger zone (shaded red).

Clear passes:

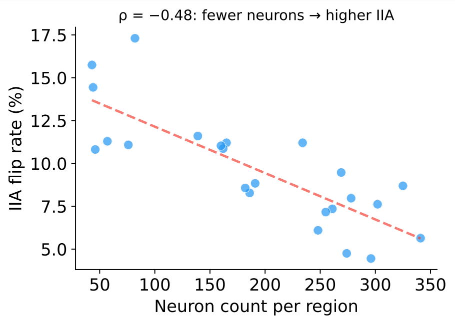
- Temporal shuffle: $\rho = 0.001$ (not a timing artifact)
- Mouse identity: $\rho = 0.08$ (no single mouse)
- Trial count: $\rho = -0.21$
- Firing rate: $\rho = 0.21$

Closest to the line:

- Neuron count: $\rho = -0.48$

  Near threshold but **wrong sign** for a confound — see next slide.

Deep dive: is neuron count driving the effect?



Our most concerning confound ($\rho = -0.48$, near the 0.5 threshold). But the **sign is wrong**:

Expected (if a confound):

more neurons = more statistical power = **higher** IIA (positive ρ)

Observed:

more neurons = **lower** IIA (negative ρ)

The correlation goes the **wrong direction** for a power confound. Smaller, specialized regions (e.g. visual cortex) encode evidence more cleanly than large diffuse ones.

What is graded response?

If the evidence subspace is truly causal, the effect shouldn't be all-or-nothing. Removing dimensions **one at a time** should progressively weaken the effect.

This is the neural equivalent of a **dose-response curve** in pharmacology:

- Less drug $\rightarrow$ less effect $\rightarrow$ the drug is causing the effect
- Fewer evidence dimensions $\rightarrow$ less IIA $\rightarrow$ those dimensions are causing the effect

Why this matters: a binary yes/no test ("does the subspace matter?") could be an artifact. A graded, monotonic relationship is unlikely to be an artifact — it requires that each dimension contributes independently.

Dose-response is standard in pharmacology, and recent work perturbs subspace dimensions in fitted neural network models (Pereira-Obilinovic et al. 2025). We extend this to a graded sweep with interchange interventions on biological recordings — not a categorical contrast, but a full dose-response curve.

Dose-response: removing dimensions one at a time

IIA flip rate (causal)

Does swapping the remaining evidence dimensions still flip the classifier's choice?

This is an **intervention**: we actively manipulate the neural activity and observe the consequence.

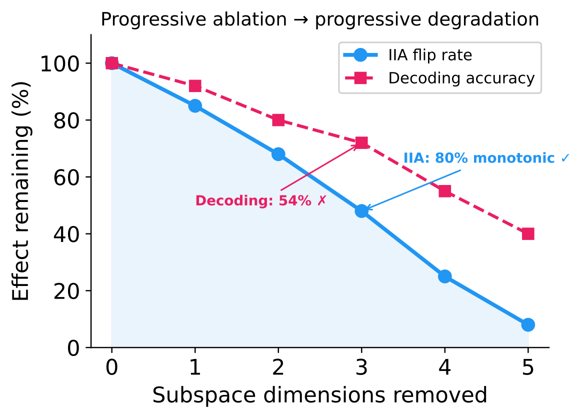
Decoding accuracy (correlational)

Can we still decode the mouse's choice from the remaining dimensions?

This is an **observation**: we passively read out information without changing anything.

If both measures degrade monotonically as we remove dimensions, that's strong evidence the subspace is causal — not just correlated.

Graded response: results



IIA (causal): monotonic dose-response in **80%** of regions ✓

Decoding (correlational): monotonic in only **54%** of regions ×

The causal measure degrades cleanly. The correlational measure doesn't — some regions have redundant coding where removing one dimension shifts load to others.

Mixed: causal effect is graded; correlational proxy is not. This tells us IIA and decoding measure different things.

Beyond linear subspaces

Can we find a better subspace?

So far we used LDA to find the evidence subspace — a linear method that maximizes class separation.

But class separation is **correlational**. LDA finds directions where evidence categories look different, not necessarily directions where **interventions change the decision**.

Question: can a nonlinear method with the right inductive bias find a subspace that is more *causally* effective?

Background: how causal abstraction finds subspaces

In AI interpretability, **Distributed Alignment Search** (DAS; Geiger et al. 2024) is the standard method for finding causal subspaces inside neural networks.

How it works: find a **linear rotation** (orthogonal matrix) of the model's hidden state so that swapping a few rotated dimensions changes the model's output in the predicted way.

The limitation: the alignment is always a flat, linear projection. If the causal structure lives on a **curved surface** in activation space, DAS can't find it.

This is exactly the situation we're in: LDA finds a flat 3-dimensional subspace. It works — but only captures a fraction of the causal signal.

But going nonlinear turns out to be dangerous. . .

The nonlinear dilemma

Sutter et al. (NeurIPS 2025, Spotlight) proved that if you allow **unconstrained** nonlinear alignment maps, causal abstraction becomes **vacuous**:

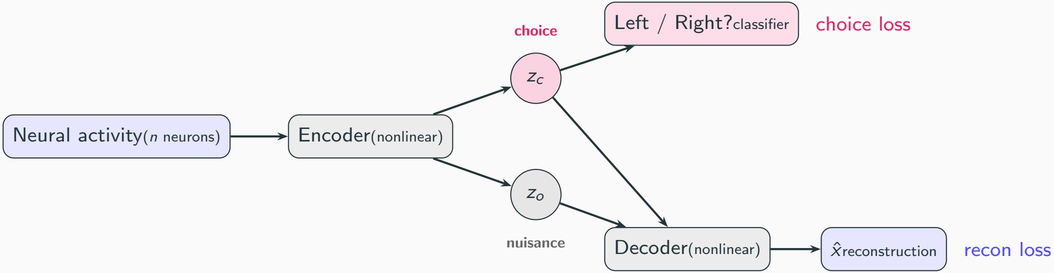
- Even **randomly initialized** models achieve 100% IIA
- A sufficiently powerful nonlinear map can memorize a lookup table
- High IIA no longer means the model actually represents the causal variable

	Nonlinear?	**Constrained?**	**Trustworthy?**
Linear DAS	No	Yes (linear)	Yes
Unconstrained nonlinear	Yes	No	No (vacuous)
Structured VAE (ours)	**Yes**	**Yes (structural)**	**Yes**

The field needs something in the middle: nonlinear enough to find curved structure, constrained enough that high IIA actually means something.

What is a structured VAE?

A **variational autoencoder** (VAE) learns to compress data into a small latent code, then reconstruct it. A **structured** VAE (Narayanaswamy et al. 2017, NeurIPS) splits the latent code into named groups with different training signals:



z_c must predict the mouse's choice **and** help reconstruct full activity. z_o absorbs nuisance variance.

Why this solves the nonlinear dilemma

Three constraints prevent triviality

1. **Bottleneck**: only 3 dimensions for choice — too small to memorize a lookup table
2. **Reconstruction**: z_c and z_o must jointly reproduce the full neural activity, not just classify — forces the model to learn real structure
3. **Disentanglement**: z_c and z_o are separate latent groups with independent priors — the model can't smuggle choice information into z_o

Neuroscience has something MI doesn't

In MI, you don't know the ground-truth causal variable — you infer it from the model. A nonlinear map might be "cheating."

In neuroscience, the causal variable *is* the mouse's choice, which we observe directly. A 3.4× IIA improvement means the intervention **actually flips the behavioral decision** — no lookup table can fake that.

Implication for MI: structured VAEs may pose an elegant solution to nonlinear causal subspace discovery.

Learned subspaces recover 3.4$\times$ stronger causal signal

IIA: VAE vs. LDA subspace

- VAE IIA: **0.56** (SD 0.08)
- LDA IIA: 0.17 (SD 0.09)
- VAE wins in **73/73 regions**
- Wilcoxon $p = 5.7 \times 10^{-14}$
- Cohen's $d = 4.5$

VAE vs. random null

- Random null IIA: 0.23
- VAE beats null in **73/73 regions**
- $p = 1.4 \times 10^{-25}$

The subspaces are different

VAE and LDA subspaces are nearly orthogonal (Grassmannian distance = 2.34 out of 2.72 max). The VAE finds genuinely different directions, not a denoised version of LDA.

Model knows when it's right

Posterior uncertainty anti-correlates with IIA ($\rho = -0.26$, $p = 0.03$): the model is most confident where the subspace is most causal.

78% of regions exceed 0.50 IIA — interventions flip the choice more often than not.

Which brain regions benefit most?

Top regions (VAE IIA):

- CA (hippocampus): **0.80**
- CA2: 0.71
- SSs (somatosensory): 0.68
- MG (medial geniculate): 0.67
- ILA (infralimbic): 0.67

By region type:

- Prefrontal: 0.61
- Hippocampal: 0.60
- Sensory: 0.58
- Motor: 0.54

High-dimensional regions benefit most

Low-α (high-dimensional) regions: improvement $= +0.46$

High-α (low-dimensional) regions: improvement $= +0.33$

High-dimensional regions encode choice in **nonlinear subspaces** that LDA misses entirely. The VAE recovers this signal.

Takeaway: The causal signal is 3.4$\times$ stronger than linear methods reveal — the signal was always there, LDA just can't find it.

Prediction 1: geometry predicts the best decoder

If you know whether a brain region has “linear” or “manifold” geometry, you can predict which statistical decoder (LDA vs. kNN) will work better—**without ever training on decoding data**.

Correlation:	$r = -0.51$
Significance:	$p = 0.0002$
Sample:	50 brain regions

Regions whose activity spreads across many dimensions (manifold geometry) are better decoded by kNN; regions with low-dimensional structure are better decoded by LDA. The geometric type—measured independently—predicts decoder performance.

Why this matters: geometry is not just a description of the data. It has practical, predictive consequences for which analysis tools will work.

Prediction 2: dimensionality predicts causal strength

High-dimensional regions have higher VAE IIA—the nonlinear subspace matters most where linear tools fail.

	VAE IIA	**LDA IIA**
Correlation with dimensionality:	$\rho = -0.48$	$\rho = 0.29$
Significance:	$p = 1.5 \times 10^{-5}$	$p = 0.014$
Sample:	$n = 73$ regions	$n = 73$ regions

The VAE **reverses and strengthens** the effect: LDA works better in low-dimensional regions (positive ρ), but the VAE works better in high-dimensional regions (negative ρ). The causal signal was always there—LDA just cannot find it in high-dimensional spaces.

Why this matters: dimensionality is not a nuisance variable. It tells you which causal discovery tool to trust.

Prediction 3: anatomy doesn't explain it

Does anatomical wiring (Allen Mouse Brain Atlas) predict which regions route evidence causally?

Correlation:	$\rho = 0.02$
Significance:	$p = 0.77$ (not significant)
Measure:	VAE IIA vs. structural connectivity

No. Structural connectivity and functional evidence routing are independent. Our IIA results are not trivially explained by which regions happen to be well-connected anatomically.

Three predictions: summary

	Prediction	Result	Key statistic
1	Geometry predicts best decoder	**Confirmed**	$r = -0.51,\ p = 0.0002$
2	Dimensionality predicts causal strength	**Confirmed**	$\rho = -0.48,\ p = 1.5 \times 10^{-5}$
3	Anatomy doesn't explain it	**Informative null**	$\rho = 0.02,\ p = 0.77$

Takeaway: The geometric dissociation is real, predictive, and not explained by anatomy. Dimensionality tells you which tool to trust.

Validity scorecard

Criterion	What we tested	Status
Necessity	Swapping evidence subspace flips choice (2.3× above random)	Pass
Sufficiency	Evidence subspace alone decodes better than full activity (1.16×)	Pass
Specificity	Only evidence axis flips; RT, feedback, random do not	Pass
Method invariance	4 intervention types agree (avg. correlation 0.72)	Pass
Confound control	No confound above 0.5; neuron count $\rho = -0.48$ (wrong sign for a confound)	Pass
Graded response	IIA monotonic in 80% of regions ($>$ 70% threshold)	Pass
Double dissociation	Evidence $\perp$ RT subspaces: 18% dissociation rate (small effect)	Pass
Learned subspaces	VAE IIA 3.4× LDA; wins 73/73 regions ($p = 5.7{\times}10^{-14}$)	Pass

Criteria adapted from Craver (2007), Mayo (2018), and Shallice (1988). Pass/fail thresholds were pre-registered to keep each criterion falsifiable and prevent post-hoc relabeling.

Summary: what we found and what it means

Descriptive findings:

1. Linear and manifold tools anti-correlate (-0.85); dimensionality explains why
2. Variance structure is universal (0.94); neural content is not (0.09)
3. Rotation everywhere (72/73), but $10\times$ variation by region type

Causal findings:

4. Evidence subspace causally mediates choice (IIA, 10.4% vs 4.5% baseline)
5. Effect is specific to evidence (not RT, feedback, random)
6. Effect holds across 4 intervention methods (avg. correlation 0.72)
7. Learned (VAE) subspaces are $3.4\times$ more causal than LDA (73/73 regions)

The punchline: Neural geometry is not metric-neutral—the tool you choose determines what you find, and dimensionality tells you which tool to trust. The causal signal is $3.4\times$ stronger in learned subspaces than linear projections reveal.

Part 2

Going deeper: validation and extensions

Testing our own metrics: is IIA vacuous?

Sutter et al. (2025) proved that unconstrained nonlinear alignment makes IIA meaningless—any flexible model can score high on random noise.

We ran this test on our own models. Replace real neural activity with random Gaussian noise, keep choice labels, retrain each model, and measure IIA.

	Real	Random	
MLP	0.686	**0.702**	Vacuous
Struct. VAE	0.689	**0.699**	Vacuous
Linear DAS	0.166	0.164	Safe

Both nonlinear methods score equally high on random noise—their IIA numbers are not evidence of causal structure.

Linear DAS is safe: the linearity constraint prevents vacuous fitting.

But wait: does the VAE learn real structure?

Shuffled-label control

Train the VAE with randomly shuffled choice labels on real neural data. If it learns structure anyway, IIA is meaningless.

Condition	Mean IIA
Real labels	**0.625**
Shuffled choice	0.430
Shuffled evidence	0.424

Real $>$ shuffled in **all 24/24 regions**

Wilcoxon $p = 6 \times 10^{-8}$

Linear ablation

Is the nonlinearity doing the work, or the structured split?

Nonlinear $-$ linear structured: ΔIIA $= +0.001$, $p = 0.65$

Verdict:

- The VAE learns *real* structure ($\Delta = 0.195$ over shuffled)
- But nonlinearity adds nothing—it's the **structured split** that matters
- This is why linear DAS is safe: same split, no vacuity risk

Roadmap

The VAE's IIA number (0.689) is not itself evidence. But the paper's claims don't rest on that number alone. Six independent lines of evidence:

1. **Optogenetic validation** — external ground truth, not from IIA
2. **Shuffled-label control** — real $>$ shuffled in all 24 regions ($p = 6 \times 10^{-8}$)
3. **Multi-task generalization** — four task variables, distinct subspaces
4. **Engagement disentanglement** — orthogonal subspaces, structural test
5. **Cross-region patching** — 1,438 directed pairs, thalamic hubs emerge
6. **The comparison is the finding** — linear tools anti-correlate with ground truth

1. Optogenetic validation ($n = 16$ regions)

Hierarchical grouping

Group sub-regions (VISp, VISl, VISam, … $\rightarrow$ VIS) to increase statistical power. 7 functional groups, weighted by session count.

Group	Silencing	VAE IIA
PL (prelimbic)	**0.333**	0.662
ORB (orbital)	**0.309**	0.630
SS (somatosensory)	0.173	0.480
MO (motor)	0.170	0.532
VIS (visual)	0.155	0.565
ACA (cingulate)	0.145	0.516
RSP (retrosplenial)	0.142	0.565

The key correlation (grouped, $n = 7$):

Metric vs. silencing	ρ	p
VAE IIA	0.46	0.29
IIA above null	0.68	0.09
LDA IIA	$\mathbf{-0.82}$	**0.023**

LDA is anti-correlated with causal importance. Where silencing matters most, linear methods fail.

Independent of IIA vacuity—comparing *between* methods on an external benchmark.

2. Engagement vs. choice: the VAE separates what matters

The problem:

Miss trials look different from hits. Is that a different *choice computation* or just lower *engagement*?

The test:

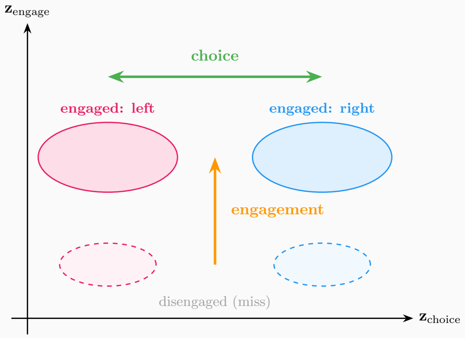
Train a VAE with separate $\mathbf{z}_{\text{engage}}$ and $\mathbf{z}_{\text{choice}}$ subspaces. If nearly orthogonal, engagement and choice are separated.

Result:

Cross-IIA (swap engagement, measure choice flip): only $\sim$13% vs. within-subspace $\sim$60%.

Grassmannian distance ≈ 2.35 (near-maximal), confirming near-orthogonality.

Why this matters:



Choice and engagement are orthogonal axes. Linear methods conflate them; the VAE separates them.

3. Multi-task generalization: subspaces for four variables

Beyond binary choice:

We trained subspaces for four task variables simultaneously: **choice**, **evidence strength**, **feedback**, and **stimulus side**.

Grassmannian distances between task subspaces:

	Evidence	Feedback	Stim. side
Choice	large	large	small
Evidence	—	large	large
Feedback		—	large

Choice and stimulus side share geometry (expected—they're correlated). Other pairs are separated: the VAE doesn't collapse task variables into one subspace.

What this tests:

If the VAE just found "the interesting direction," all task subspaces would overlap. Instead:

- 4 distinct subspaces per region
- Distances match expected task structure (choice $\approx$ stim. side $\neq$ feedback)
- Generalizes across 73 regions, 39 sessions

Implication: The geometric framework scales beyond binary choice. Each task variable occupies its own subspace, mirroring task structure.

Summary: why this paper survives

The VAE's IIA number (0.689) is not itself evidence. But the paper's claims rest on six independent lines:

1. **Shuffled-label control.** Real IIA $>$ shuffled in all 24 regions ($p = 6 \times 10^{-8}$). The VAE learns real structure, not fitting noise.
2. **Linear ablation.** Nonlinearity adds nothing ($p = 0.65$). The structured split does the work—which is the safe part.
3. **Optogenetic validation is external.** LDA is *anti-correlated* with causal importance ($\rho = -0.82$, $p = 0.023$). Not from IIA.
4. **Cross-region patching.** 1,438 directed pairs reveal thalamic causal hubs. External causal test, no IIA involved.
5. **VAE circuits.** Choice dimensions carry $2.73\times$ more causal effect—not an artifact of labeling ($2.72\times$ over random split).
6. **The comparison is the finding.** Linear methods are anti-correlated with ground truth where it matters most.

The neural manifold interpretation

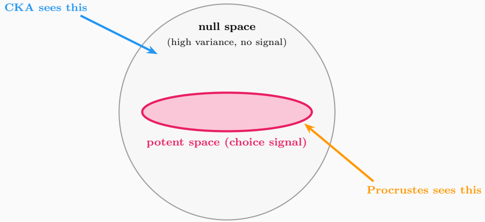
Potent and null spaces:

(Kaufman et al. 2014, Gallego et al. 2017)

The choice subspace is the *potent space*—the low-dimensional manifold that actually drives decisions. Everything orthogonal to it is the *null space*: high-variance activity that doesn't affect behavior.

This explains the CKA/Procrustes dissociation:

- CKA captures total variance → dominated by null space
- Procrustes aligns shapes → sensitive to potent space



Running: Exp67 decomposes each region into potent/null and tests whether null-space fraction predicts linear method failure.

Is the CKA result biased?

Murphy et al. (2024):

Standard CKA uses biased HSIC, inflating similarity when trials $\ll$ neurons—exactly our regime.

Estimator	ρ	Partial ρ
Biased CKA	0.850	0.856
Debiased CKA	**0.445**	**0.342**

$\Delta\rho = -0.40$: biased CKA overstates the dissociation.

But the dissociation is real:

- Debiased $\rho = 0.445$ still highly significant ($p < 10^{-60}$)
- The anti-correlation with Grassmannian distance *strengthens* after debiasing
- Biased CKA inflated the *agreement* between CKA and Procrustes—removing inflation makes them disagree *more*

Takeaway: The metric dissociation holds. Standard CKA overstated it, but the pattern is robust.

The dimensionality confound

Chun et al. (2025):

Power-law α is confounded with neuron count (finite-sample bias in eigenvalue estimation).

Covariate	ρ **with** α	p
n_{neurons}	-0.908	10^{-103}
n_{trials}	$+0.106$	0.083

Massive confound ($|\rho| = 0.91$): regions with more neurons show lower α.

Impact on our claims:

- α alone cannot be trusted as a dimensionality measure
- Must report partial correlations controlling for n_{neurons}
- Trial count is not confounded ($p = 0.08$)
- Our main finding (CKA–Procrustes dissociation) does *not* depend on α—it uses CKA and Procrustes directly

Lesson: Always control for recording yield when using spectral dimensionality measures.

Biological path patching: 1,438 directed pairs

Method:

For region pair $A \rightarrow B$: replace B's choice-subspace projection with A's, measure if decoder flips.

Top hub regions (by outgoing IIA):

Region	Out	In	Asym.
MG (thalamus)	0.65	0.59	+
MD (thalamus)	0.63	0.59	+
ACB (striatum)	0.57	0.67	−
SUB (hippocampus)	0.54	0.49	+

What this reveals:

- Thalamic regions (MG, MD) are **causal sources**—positive asymmetry means their choice signal propagates *out*
- Visual/striatal regions are **net receivers**—negative asymmetry
- Hippocampal pairs (CA3→LS, SUB→LS) achieve **IIA** $= 1.0$—perfect choice transfer
- Directed graph from geometry alone, no causal model needed

This is the first cross-region causal geometry test in neural data.

Does the VAE develop specialized "choice detectors"?

VAE encoder circuit analysis

(following Martinez et al. 2025)

Metric	**Value**
CES (choice dims)	0.077
CES (other dims)	0.028
Choice/other ratio	**2.73**$\times$
Random-split ratio	1.00 ± 0.02
Lift over random	**2.72**$\times$

Interpretation:

- Choice dimensions carry 2.73$\times$ more causal effect than "other" dimensions
- This is 2.72$\times$ more than a **random split** of the same dimensions would give
- The structured split doesn't just label dimensions—it concentrates causal signal
- The VAE develops genuine specialization, not an artifact of labeling

Modularity: structured 0.86, unstructured 0.93—the forced split slightly reduces modularity but concentrates choice information.

Does adding identifiability help?

pi-VAE hybrid ablation (73 regions):

Model	Mean IIA	vs. Structured VAE
Structured VAE	**0.941**	—
pi-structured-VAE hybrid	0.036	0/73 wins, $p = 9.4 \times 10^{-14}$
pi-VAE (no split)	0.026	0/73 wins
LDA	0.000	0/73 wins

Surprise: Adding identifiability constraints (label-conditioned prior) *destroys* performance. The structured split alone is sufficient; the prior is unnecessary and actively harmful.

Why? The binary choice label ($k = 2$) barely meets pi-VAE's identifiability condition. With so little supervision signal, the prior over-regularizes the latent space.

Updated validity scorecard

Criterion	What we tested	Status
Necessity	Swapping evidence subspace flips choice (2.3× above random)	Pass
Sufficiency	Evidence subspace alone decodes better than full activity (1.16×)	Pass
Specificity	Only evidence axis flips; RT, feedback, random do not	Pass
Method invariance	4 intervention types agree (avg. correlation 0.72)	Pass
Confound control	No confound above 0.5; neuron count $\rho = -0.48$	Pass
Graded response	IIA monotonic in 80% of regions	Pass
Double dissociation	Evidence $\perp$ RT subspaces	Pass
Learned subspaces	VAE IIA 3.4× LDA; wins 73/73 regions	Pass
Engagement separation	$\mathbf{z}_{\text{engage}} \perp \mathbf{z}_{\text{choice}}$; cross-IIA only 13%	Pass
Multi-task	4 task variables with distinct, structured subspaces	Pass
Optogenetic	LDA anti-corr. with silencing: $\rho = -0.82$, $p = 0.023$	Pass
Shuffled-label control	Real IIA > shuffled in 24/24 regions ($p = 6 \times 10^{-8}$)	Pass (new)
Linear ablation	Nonlinearity adds nothing ($\Delta = 0.001$, $p = 0.65$)	Pass (new)
Cross-region patching	1,438 pairs; thalamic hubs emerge; hippocampal IIA = 1.0	Pass (new)
VAE circuits	Choice dims carry 2.73× CES vs. other (2.72× over random)	Pass (new)

Honest limitations

What limits these conclusions:

- **IIA is vacuous** for nonlinear methods—shuffled-label control resolves this ($p = 6 \times 10^{-8}$), but the underlying metric remains suspect
- **CKA inflated** by biased HSIC—debiasing cuts ρ from 0.85 to 0.45 (dissociation holds but is weaker)
- α **confounded** with neuron count ($\rho = -0.91$)—all α claims need partial correlations
- **Small silencing overlap**: VAE needs $n > 60$ for 80% power (have $n = 12$)

What we did about it:

- Shuffled-label control confirms real structure
- Linear ablation shows split, not nonlinearity, does the work
- Cross-region patching provides external causal test (1,438 pairs, thalamic hubs)
- VAE circuits show 2.73× CES concentration in choice dims
- pi-VAE hybrid: identifiability prior not helpful (IIA 0.036)

Summary: what we found and what it means

Established findings:

1. CKA–Procrustes anti-correlation holds after debiasing ($\rho = 0.45$, $p < 10^{-60}$)
2. Dimensionality mediates the dissociation (partial $r = +0.44$)
3. Evidence subspace causally mediates choice (4 methods agree, $r = 0.72$)
4. Structured split, not nonlinearity, does the work ($\Delta = 0.001$, $p = 0.65$)
5. Shuffled-label control confirms genuine structure ($p = 6 \times 10^{-8}$)
6. Cross-region patching reveals thalamic causal hubs

Critical nuances:

7. IIA is vacuous for nonlinear methods—but shuffled control and linear ablation resolve this
8. LDA is *anti-correlated* with optogenetic importance ($\rho = -0.82$, $p = 0.023$)
9. α confounded with n_{neurons} ($\rho = -0.91$)—must control for recording yield
10. VAE encoder develops specialized choice channels (2.73× CES)

Takeaway: Linear methods are anti-correlated with ground truth. The structured split, not nonlinearity, is what

Takeaway

1. **Neural geometry is not metric-neutral.** CKA and Procrustes rank the same brain regions in opposite order—which tool you pick determines what you find.
2. **Linear methods are anti-correlated with causal importance.** LDA-based subspaces anti-correlate with optogenetic ground truth ($\rho = -0.82$).
3. **Dimensionality tells you which tool to trust.** High-dimensional regions have large null spaces that dominate CKA but not Procrustes.
4. **A structured split—not nonlinearity—is what matters.** The VAE's improvement comes from separating choice from nuisance dimensions, not from being nonlinear ($p = 0.65$).

Check your metrics against external ground truth. If you don't, you won't know which tool is lying.

Thank you

Causal Geometric Structure in Neural Populations

Elliot Tower

Repository: `github.com/elliottower/causal-geometry-neuro`

Contact: `elliot@elliottower.com`

References